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From Guidance to Surveillance:

An Ethical Framework for Analyzing Data Collection Policies and Practices in Google Maps

Case Study Report

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Abstract-Google Maps is one of the most widely used navigation tools, relying on GPS, user history, and real-time data to provide efficient routing. However, these features depend on continuous tracking, raising concerns about privacy, transparency, and user autonomy. This study uses a qualitative, literature-based approach to examine key ethical issues in Google Maps, focusing on surveillance risks, weak informed consent, and cognitive dependence on GPS. Findings show that many users are unaware of how their location data is stored or reused, and trust decreases when platforms lack clear communication about data practices. Evidence also suggests that heavy reliance on GPS may weaken spatial memory. Overall, the study emphasizes the need for greater transparency, stronger consent mechanisms, and responsible data governance to protect user rights.

Keywords: Privacy, Surveillance, Informed Consent, Data Governance, GPS Dependence, Digital Ethics, Google Maps, Spatial Memory.

I. Introduction

Google Maps is a great location-tracking and navigation application all over the world that its users in getting quality geographical information for daily destinations, such as malls and work places with information about the past or present state of a visited site is also available while staying updated on traffic conditions. Even if it makes life easier it also collects data permanently, this raising an important concerns about privacy and surveillance. Many users are unaware of how much data is collected or how it is used. Also, constant and frequent reliance on using the application, weaken natural navigation skills and memory. All these issues matter because they affect trust, independence, and daily life. Therefore, in this paper, we explore the following questions:

1. How do Google Maps data collection and navigation practices raise ethical concerns around

privacy, surveillance, and everyday human reliance on technology?

2. How does the perception of being constantly tracked by Google Maps affect public trust in Google and similar technology companies that provide digital services?

3. To what extent does Google Maps provide users with meaningful informed consent about how their location data is collected, stored, and shared?

4. What is the impact of frequent reliance on Google Maps on user's cognitive abilities and everyday navigation?

II. Literature Review

Research on navigation technologies such as Google Maps highlights three main ethical concerns: privacy, surveillance, and cognitive dependence. Several studies show that users often lack awareness and control over how their personal data is collected and used. Alashwali and Cranor [1] report that many Google account holders feel uneasy about Google's data practices, reflecting weak informed consent and limited transparency. Similar findings appear in Raibulet and Wang [2], who show that most users do not fully understand Google Maps' privacy policy or how location data is processed. Fox et al. [3] further emphasize that unclear data practices reduce user trust across digital services.

Sieg [4] adds that mobile location data can easily be repurposed without user knowledge, increasing the risk of misuse. Savino et al. [5] support this by showing that map applications collect detailed behavioral patterns during real-world use. Studies in digital surveillance also highlight trust issues: Shaghaghi et al. [6] and Toch and Ayalon [7] found that even the perception of being monitored negatively affects user confidence and autonomy, aligning with broader frameworks on surveillance ethics. Macnish and Henschke [8] further argue that continuous tracking can change user behavior and create long-term feelings of exposure.

Another important theme is the cognitive impact of GPS dependence. Dahmani and Bohbot [9] show that frequent GPS use can weaken spatial memory, while Yan et al. [10] found that reliance on navigation apps reduces individuals' natural sense of direction. Topete et al. [11] further link heavy GPS use to increased spatial anxiety. Prestby [12] notes that excessive trust in digital maps can also reduce users' ability to critically evaluate route information. Overall, the literature indicates that while Google Maps provides convenience and efficiency, it raises ethical challenges related to privacy, trust, and long-term cognitive well-being.

III. Methodology

This case study uses a qualitative research approach based mainly on a literature review. The study relies on information gathered from research papers, academic articles, and books that discuss Google Maps, data surveillance, privacy, and the growing dependence on GPS technology.

The collected data is analyzed through thematic and ethical analysis, focusing on key themes such as user surveillance, data transparency, consent, and how society increasingly depends on user-generated GPS data. The analysis examines different authors' perspectives and evaluates how Google Maps balances technological innovation with user privacy and reliability in location-based information.

IV. Case Description

Google Maps is a navigation system that helps people reach places easily, like homes, schools, shops, and hospitals, It works by collecting GPS signals and online data to understand where the user is and suggest the fastest or easiest route, The system was developed by Google company and has become a daily tool for millions of users because it makes moving around more efficient. The system gathers different types of information, like the user's current location, places they visited before, search history, and time of travel. Then it uses AI algorithms to analyze these details,

predict traffic, calculate arrival times, and recommend personalized routes. While these features improve accuracy and convenience, they also depend on collecting sensitive data, which raises concerns about privacy and surveillance. Several groups are involved in and affected by Google Maps, Users rely on it for everyday navigation, but face risks related to data collection, Google develops the system and uses the data to improve its services, Governments are responsible for ensuring data protection laws are followed, finally we have Local communities and businesses are also affected cause the routes suggested by the app can change how cars move in an area and affect how easily people can find certain stores.

V. Ethical Analysis

i. Privacy at a Crossroads: Risks in Google Maps Data Practices

Google's collection of personal data raises a serious ethical concern: user privacy. Personal information is gathered continuously, often without full understanding or genuine consent, putting user autonomy and trust at risk. Building on this, the continuous collection exposes users to real harm, as their sensitive data can be exploited for targeted advertising or surveillance. This lack of informed consent not only threatens privacy but also undermines confidence in digital services, leaving users vulnerable and powerless over their own information. This is clearly problematic because they violate users' fundamental right to privacy. They undermine trust, treat individuals as mere data sources, and can harm vulnerable groups, highlighting the importance of balancing technological benefits with ethical responsibilities. Research by Alashwali and Cranor [1] shows that Google account holders in Saudi Arabia often feel uneasy about how their personal data is collected and used, highlighting real concerns over privacy and informed consent. Additionally, Sieg [4] points out that mapping mobile phone location data raises ethical challenges, as it can be used for undisclosed purposes, further emphasizing violations of privacy and trust. This highlights the

urgent need for technology companies like Google to implement ethical data practices that respect user privacy, ensure informed consent, and maintain trust while still delivering technological benefits.

ii. Living Under the Digital Gaze: Surveillance and Public Trust

Google Maps constantly tracks users in the background, creating a form of continuous surveillance where people benefit from the app but are also being closely observed. Each movement such as daily commutes or visits to private places like hospitals or mosques can be recorded and stored, forming a digital trail that extends far beyond what is needed for navigation. Studies by Shaghaghi et al. [6] and Toch and Ayalon [7] found that even the *feeling* of being watched reduces users' trust in technology companies. This reflects Foucault's "panoptic effect," where people change their behavior because they believe they are being monitored. From a deontological perspective, Google has a moral duty to respect users' privacy and treat them as individuals, not just as data sources. The justice view also shows that constant tracking can harm certain groups who fear their information might be exposed or misused. Some might argue that tracking brings benefits like smoother traffic or faster navigation but these advantages do not make up for the loss of personal privacy and trust. To maintain public confidence, Google should be transparent, collect only what is necessary, and clearly limit how personal data is used.

iii. Informed Consent and Transparency in Location Data Use

The use of location data raises key ethical issues concerning the absence of informed consent and transparency. Many users are unaware that navigation apps continuously collect and share their movements, revealing personal patterns such as home or workplace locations. This constriction of openness prevents individuals from making real decisions about their privacy and Independence, given that these decisions are fundamental rights for users. The risks of consent and limited transparency have

double effect on both individuals and communities. Location data can expose sensitive behaviors like visits to hospitals or religious sites, which may be misused without permission. On a larger scale, navigation systems can redirect traffic into quiet neighborhoods, generating problem related to safety and environmental that citizens never agreed to. From an ethical view, utilitarian and deontological frameworks explain why this problem matters. While utilitarianism provide public benefits such as smooth traffic, but these are not a sufficient justification to hidden risks or use it inappropriate way . A deontological perspective stresses that people must be treated as moral agents with rights, not just data sources. Research supports these concerns. Dahmani and Bohbot [9] found that habitual GPS use weakens spatial memory, showing that technology can affect human cognition. These findings confirm that transparency is not only a legal duty but a moral responsibility. In conclusion, ethical data use requires clear transparent communication genuine consent. Companies should explain how and when data are collected and shared, allow users to limit or delete their information, and commit to transparency as the foundation of digital trust.

iv. Outsourcing Our Minds: Cognitive Impact Of GPS Reliance

The main ethical issue surrounding reliance on GPS technology is that constant dependence on it weakens natural spatial awareness and memory, raising concerns about technological overdependence and its impact on human autonomy and cognitive growth. The harm lies in the decline of essential cognitive abilities such as spatial reasoning and problem solving, leaving people vulnerable when technology fails. Over time, this dependence can reduce confidence and engagement with the real world. From a utilitarian view, GPS systems offer safety and convenience, but from a virtue ethics or rights based view, overreliance undermines virtues like self reliance and attentiveness. Ethically, a balance must be struck between technological benefits and preserving cognitive well being. Research

supports these concerns: Dahmani and Bohbot [9] found that habitual GPS use impairs spatial memory, Yan et al. [10] reported that dependency reduces one's sense of direction, and Topete et al. [11] linked it to spatial anxiety. These findings emphasize the need for designs that support navigation without weakening human cognitive skills.

Overall, the ethical analysis shows that Google Maps presents interconnected concerns related to privacy, surveillance, informed consent, and cognitive dependence. Although the application provides significant benefits in navigation and daily efficiency, these advantages come with ethical risks that affect user autonomy and long-term trust. Strengthening transparency, improving consent mechanisms, and reducing unnecessary tracking are essential steps toward balancing technological usefulness with user protection.

VI. Findings

The findings show that Google Maps raises major ethical issues related to privacy, surveillance, and cognitive dependence. Continuous tracking and unclear consent processes violate user privacy and reduce public trust in technology. The lack of transparency in how data are collected and shared reflects broader governance problems and weak accountability in technology companies. Studies confirm [1], [6], [7] that users feel uneasy about how their information is used, and research also shows [9]-[11] that heavy reliance on GPS can weaken spatial memory and independence. Overall, while Google Maps provides convenience and efficiency, it also challenges user autonomy and trust, highlighting the need for stronger transparency, accountability, and ethical data practices.

Google's culture focuses on innovation and efficiency but often overlooks ethical accountability and user protection. The company prioritizes performance and competitiveness over transparency, leading to complex policies that limit informed consent. With little ethical oversight, user data are treated as business assets

rather than personal information. To improve, Google should adopt ethics by design principles, conduct regular privacy audits, and build a culture of transparency and accountability.

In summary, the findings demonstrate that although Google Maps offers convenience and efficiency, it also introduces ethical challenges involving privacy, surveillance, and cognitive reliance. Users often remain unaware of how their data is stored or used, and research consistently shows concerns related to trust and autonomy. These findings highlight the need for stronger transparency, clearer consent processes, and responsible data practices to ensure that the benefits of the technology do not compromise user rights.

VII. Recommendations

To address the privacy and surveillance concerns identified in this case, Google Maps should adopt stronger protections that limit continuous tracking, increase user control, and ensure full transparency. Technically, this includes minimizing background location collection, improving anonymization, and expanding on device processing. Clear and simple consent mechanisms, real time permission dashboards, and automatic deletion of location history would help users understand and manage their privacy. Google must also restrict secondary uses of sensitive location data and introduce independent oversight to prevent surveillance misuse. Organizationally, adopting ethics by design and issuing regular transparency reports are essential steps for rebuilding user trust and ensuring responsible data practices. A concise set of recommendations can help address ethical concerns related to location data use and GPS overreliance. To reduce GPS overdependence, systems can include a “Learn the Route” mode that encourages users to navigate familiar trips with reduced guidance, as well as a “Safe Exploration Mode” that allows short distance navigation without full instructions, intervening only when users deviate significantly. Future research should explore long term cognitive

effects and evaluate improved consent models while examining broader societal impacts, such as traffic redistribution into residential areas.

VIII. Conclusion

In this case study, we explored how Google Maps provides useful navigation services but also creates several ethical concerns related to privacy, surveillance, and dependence on GPS. The analysis showed that the app collects large amounts of location data, often without users fully realizing how their information is stored or shared. This lack of clear consent and transparency can reduce trust and make people uncomfortable about being constantly tracked. The research also showed that relying too much on GPS can weaken spatial memory and natural navigation skills over time.

Overall, while Google Maps helps people move around more easily, these benefits come with important responsibilities. Google needs to be more transparent about its data practices, offer clearer consent options, and reduce unnecessary tracking. Navigation systems should also support users in keeping some level of independence instead of encouraging complete dependence on technology.

In the end, the findings highlight that convenience should not come at the cost of user rights. Improving privacy, communication, and ethical design will help ensure that people can enjoy the advantages of Google Maps without risking their trust, independence, or long-term well-being.

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